

Article

Single-Stage CMOS Operational Transconductance Amplifiers (OTAs): A Design Tutorial

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Abstract: This paper presents a comprehensive design tutorial for four types of single-stage operational transconductance amplifiers (OTAs): (1) five-transistor OTAs, (2) telescopic cascode OTAs, (3) folded cascode OTAs, and (4) current mirror OTAs. These OTAs serve as fundamental building blocks in analog circuits. The operational principles of each OTA are reviewed, and the key performance metrics are derived through a hand analysis. These performance metrics encompass most crucial parameters, including small-signal parameters, frequency response, input and output swing ranges, rising and falling slew rates, nonidealities, and bias circuit simplicity. All of these metrics are verified and compared using the simulation. Furthermore, the practical applications of each OTA are summarized, and a case study demonstrates the enhancement of a neural recording amplifier's performance through appropriate OTA selection. A thorough review of the essential building blocks will become a stepping stone to design high-performance analog amplifiers across diverse applications.

Keywords: analog amplifiers; single-stage OTAs; five-transistor OTA; telescopic cascode OTA; folded cascode OTA; current mirror OTA; bias circuit; swing range; small signal analysis; slew rate; circuit noise



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1. Introduction

Operational transconductance amplifiers (OTAs), which convert an input voltage to an output current, play crucial roles in a wide range of electronic systems. G_m -C filters, based on OTAs, have been employed in sensor interface integrated circuits (ICs) [1–3], biopotential recording circuits [4–6], and wireless communication integrated circuits (ICs) [7–9] to attenuate interferers with a wide tuning range of cut-off frequencies. OTA-based integrators with an inherent high gain and low-pass frequency response have been used in delta-sigma analog-to-digital converters (ADCs) [10–12]. Moreover, OTAs themselves have served as feedforward amplifiers of closed-loop systems such as capacitively coupled instrumentation amplifiers (CCIAs) [13–15] and power ICs [16–20] thanks to their high gain performance. The versatility and wide applicability of OTAs make them indispensable in various analog and mixed-signal circuit designs, emphasizing the paramount importance of OTA design.

Single-stage OTAs have gained significant interest due to their inherent simplicity and relatively lower power consumption compared to multi-stage OTAs, making them highly attractive for integration into ICs. Moreover, they can serve as the input stage of multi-stage operational amplifiers (OP-AMPs) and multi-stage OTAs. As a result, a comprehensive understanding of single-stage OTAs becomes crucial in the pursuit of high-performance analog circuit designs. This paper focuses on providing a comprehensive design tutorial for various types of single-stage OTAs, serving as a solid foundation for early-stage circuit designers.

To achieve this goal, this paper presents a comprehensive design tutorial covering four different types of single-stage OTAs: (1) five-transistor OTAs, (2) telescopic cascode OTAs, (3) folded cascode OTAs, and (4) current mirror OTAs. These OTAs are widely used and representative among various single-stage OTAs. Moreover, these four types are basic structures of single-stage OTAs that are included in the textbook [21–23]. Understanding these structures and driving performance metrics could be helpful for early-stage circuit designers to have a foundation to explore more advanced structures. For each OTA, the operation principle is reviewed, and the key performance metrics are derived through a hand analysis and then verified based on a rigorous simulation. Furthermore, the practical applications of each OTA are summarized. Finally, a case study is provided to demonstrate the tutorial's practical implementation. A suitable OTA structure is chosen based on the analytical results, thereby improving the performance of a conventional neural recording amplifier. A thorough analysis of the basic building blocks serves as a valuable resource for designing high-performance analog amplifiers across diverse applications. It is important to note that this tutorial goes beyond conventional textbooks [21–25], offering a more practical and applied approach through a comparative analysis, application discussions, and a detailed case study.

The rest of this paper is organized as follows: Section 2 introduces the background of OTAs, with a primary focus on performance metrics. Section 3 elaborates on the operational principles and performance metrics derived through a hand analysis. Section 4 presents the simulation results and comparisons. In Section 5, the applications and a case study are discussed. Finally, Section 6 concludes the paper, summarizing the findings and implications of the comprehensive OTA design tutorial.

2. Background

In this section, we review the most crucial performance metrics of OTAs that will be analyzed in depth throughout the paper. In addition, we provide a brief explanation for the motivations behind the four types of single-stage OTAs: (1) five-transistor OTAs, (2) telescopic cascode OTAs, (3) folded cascode OTAs, and (4) current mirror OTAs. These performance metrics and design motivations are fundamental for understanding the characteristics and applications of each OTA type, paving the way for a comprehensive analysis and comparisons in the subsequent sections.

Figure 1 presents the electrical performance metrics of OTAs, encompassing various crucial parameters such as small-signal parameters, frequency response, input and output swing ranges, rising and falling slew rates, circuit noise, and bias circuit simplicity. The structure of the OTA plays a significant role in determining these metrics and, consequently, affects the overall performance and behavior of the amplifier. Understanding how these parameters are influenced by the OTA's structure is vital for optimizing the design and achieving high-performance analog amplifiers across different applications.

The analysis of the small-signal parameters and frequency response is crucial in OTA design, especially considering that the input of the amplifier is typically a small AC signal. Among the small-signal parameters, two fundamental ones, effective transconductance (G_m) and output resistance (r_{out}), will be analyzed thoroughly. The G_m represents the ability to convert an input voltage to an output current, and the output resistance (r_{out}) represents the capability to deliver a constant current regardless of the load conditions. These two small-signal parameters, G_m and r_{out} , are dominant parameters that determine the frequency response of a single-stage OTA, which includes DC gain (A_o), 3-dB angular frequency (ω_{3dB}), and unity-gain angular bandwidth (ω_{GB}). Properly setting these parameters according to the desired signal bandwidth is essential for optimizing the OTA's performance in specific applications.

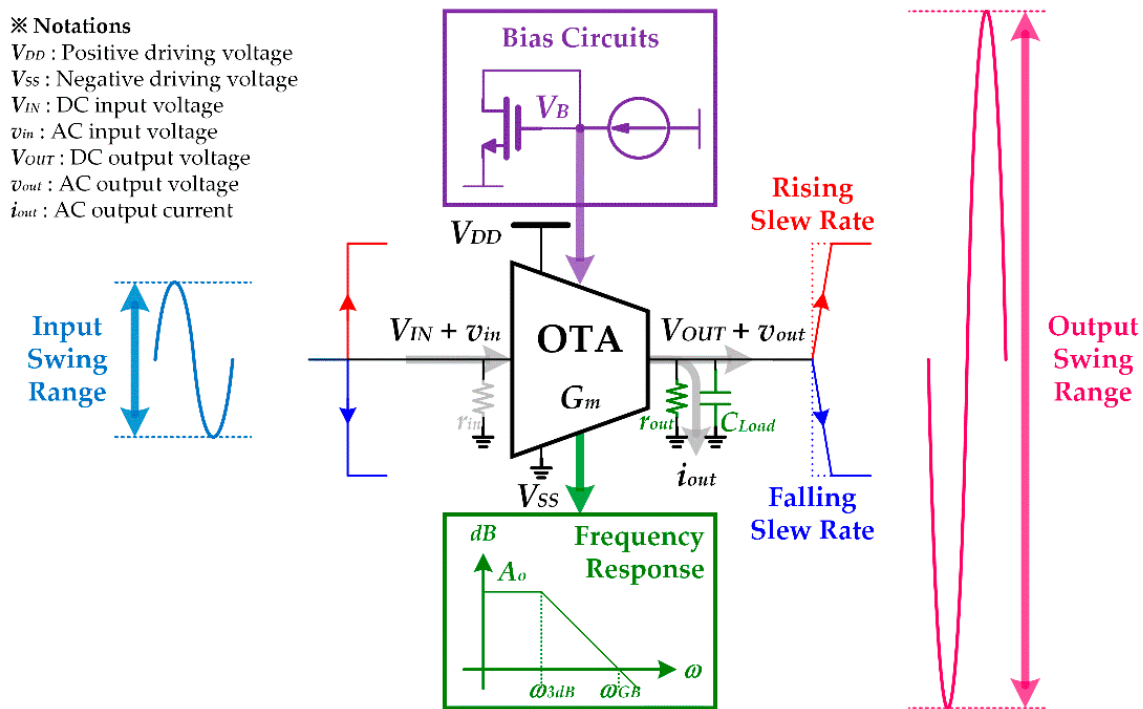


Figure 1. Performance metrics of an OTA.

The input and output swing ranges are also crucial performance metrics that determine the maximum and minimum allowable input and output voltage levels for the proper operation of the OTA. These swing ranges are particularly important when the OTA is placed in front of another amplifier, ensuring compatibility and preventing signal distortion. Having sufficient swing ranges allows for the OTA to handle a wide range of input signals without clipping or distortion, which is essential for maintaining the integrity of the amplified signal in practical applications.

Another important performance metric is the slew rate (SR). Amplifiers often handle fast input signals with frequencies higher than ω_{GB} , and the SR indicates how fast the OTA drives the output in this condition. As shown in Figure 1, there are two types of SRs: rising SR and falling SR. Steeper slopes of the output signal for rising and falling edges indicate an OTA that has a better performance.

The noise of the OTA is also a crucial metric when the input signal is small and/or the target system requires a high resolution. In particular, since the total noise of the system is dominant at the front end of the system, the OTA used for the front end should have a low noise level. Although it is impossible to eliminate the noise entirely, it is necessary to analyze the circuit noise to minimize it as much as possible. The same analysis can be extended to other circuit nonidealities such as offset and mismatches.

At last, considering that the advantage of single-stage OTAs lies in their simple structure, this simplicity should extend to the implementation of bias circuits. Bias circuits play critical roles in providing the necessary DC operating conditions for the OTA, ensuring its stable and optimal performance. However, if complex bias circuits are needed, this can compromise the benefits of using a single-stage OTA. Therefore, to fully leverage the advantages of single-stage OTAs, it is important to keep the bias circuits simple and straightforward. This will allow for the efficient and effective use of single-stage OTAs in analog circuit design.

A single transistor can be regarded as a voltage-controlled current source that has its own G_m and r_{out} . That is, for single-ended signals and a sufficiently large r_{out} , a single transistor can be used as a transconductance amplifier (TA) that converts an input voltage to an output current. However, for differential signaling, high load impedance, or high driving

capability requirements, more advanced single-stage OTAs are required. The four types of single-stage OTAs reviewed in this paper are designed with differential signaling. The telescopic cascode OTA and folded cascode OTA achieve much higher output impedance than a single transistor. A current mirror OTA is suitable for achieving a high driving capability. In Section 3, all of these OTA types will be analyzed, and their performance metrics will be evaluated.

3. Hand Analysis

This section provides a quantitative analysis of the performance metrics of four types of single-stage OTAs, as introduced in Section 2. This analysis offers valuable insights into the capabilities and limitations of each OTA type, aiding in the selection of the most suitable one for specific applications.

3.1. Five-Transistor OTA

The five-transistor OTA shown in Figure 2 is one of the most basic and widely used OTAs. The input transistors, M_1 and M_2 , form a differential transconductor. An active current mirror formed by M_3 and M_4 is loaded onto the transconductor. M_5 serves as a tail current source, providing a current to the other four transistors.

See Figure 2a to look into the design of the bias circuit and operation at equilibrium. The current source, $I_{B,5}$, is copied to M_5 via the diode-connected transistor. In general, the most clean and/or precise current, which is generated via a bandgap reference, can be copied simply by only one more diode-connected transistor. This simple bias circuit becomes one of the advantages of a five-transistor OTA compared to OTAs based on cascode structure. At the equilibrium, M_1 and M_2 generate current signals, I_{D1} and I_{D2} , respectively. Since I_{D1} is fed into M_3 , the current flowing through M_4 is same as in I_{D1} . As shown in Figure 2a, when two input signals have the same voltage level, I_{D1} and I_{D2} have the same magnitude with the same direction. I_{D1} and I_{D2} are canceled, and no current flows to the load. In contrast, when two input signals have the same DC voltage level with a small AC differential signal, v_{id} , a current flows to the load due to the different directions of I_{D1} and I_{D2} .

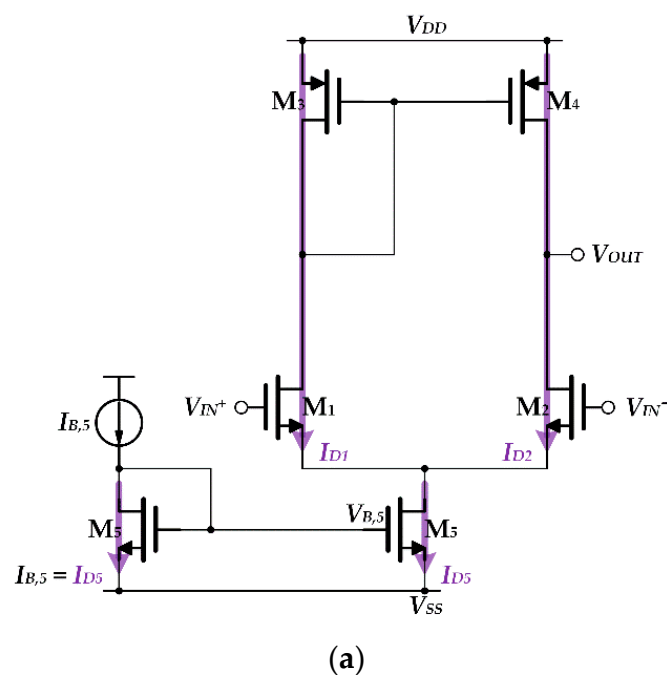


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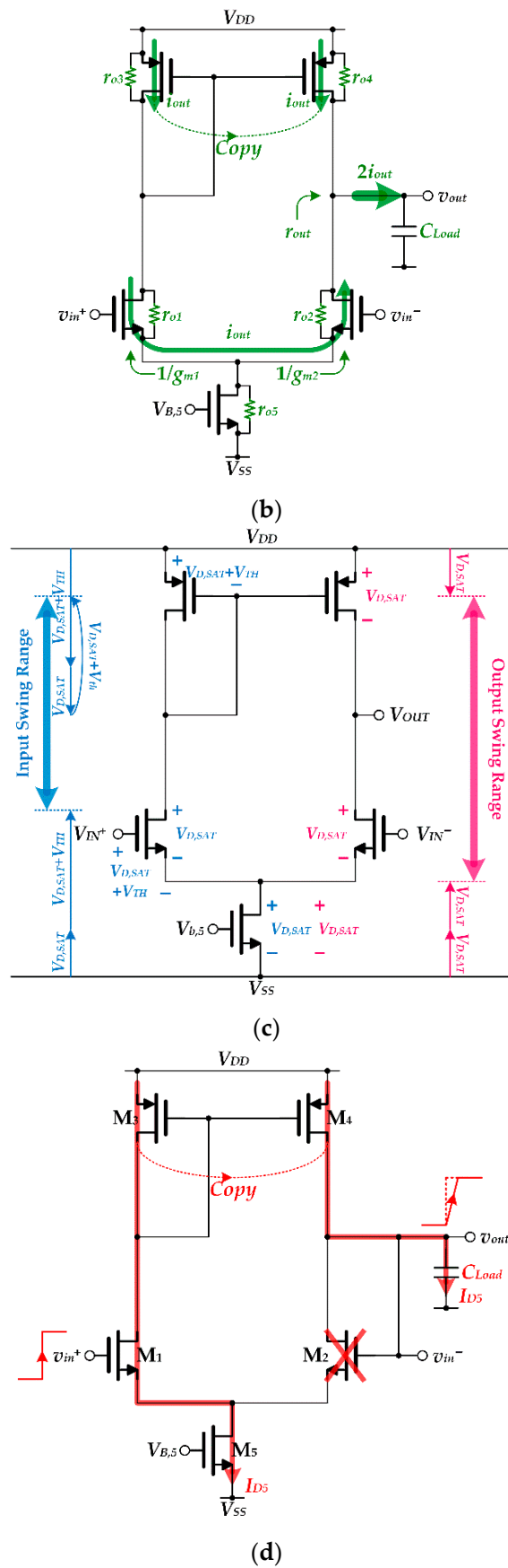


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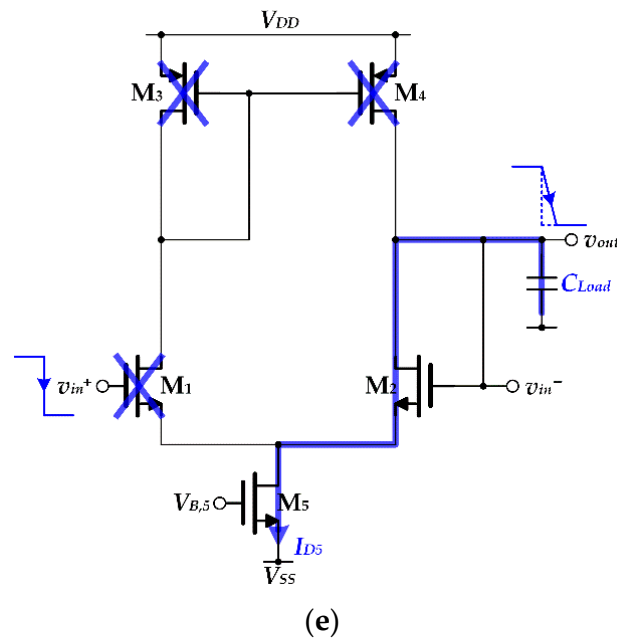


Figure 2. Five-transistor OTA for showing (a) operation at equilibrium, (b) small-signal parameters and frequency response, (c) input and output swing ranges, (d) rising SR, and (e) falling SR. In (c), $V_{D,SAT}$ is the minimum required V_{ds} for maintaining the saturation operation of transistors.

See Figure 2b for a small signal analysis with a differential input signal. When M_1 and M_2 are matched with the same transconductance, represented as g_m , M_1 and M_2 generate $i_{out} = g_m(v_{id}/2)$ and $-i_{out}$. Note that the current flowing through M_4 is also i_{out} due to the current mirror. Two currents from M_2 and M_4 are merged together at the output node, thereby flowing $2i_{out} = g_mv_{id}$ to the load. In this regard, the G_m of a five-transistor OTA, $G_{m,5TR_OTA}$, is expressed as follows:

$$G_{m,5TR_OTA} = g_{m1} = g_{m2} = g_m \quad (1)$$

where g_{m1} and g_{m2} are the transconductances of M_1 and M_2 , respectively.

Now, let us look into the r_{out} of a five-transistor OTA. At the output node, r_{o2} and r_{o4} are connected in parallel. Therefore, when M_1 and M_2 are matched with the same output impedance, represented r_o , and when r_{o2} is much higher than $1/g_{m1}$, the r_{out} of a five-transistor OTA, $r_{out,5TR_OTA}$, is expressed as follows:

$$r_{out,5TR_OTA} = r_{o2} || r_{o4} = \frac{r_o}{2} \quad (2)$$

where r_{o2} and r_{o4} are the output impedances of M_2 and M_4 , respectively. The higher the r_o , the higher the $r_{out,5TR_OTA}$. A high $r_{out,5TR_OTA}$ is required to provide a constant current regardless of the load resistance or capacitance.

Next, we analyze the frequency response of a five-transistor OTA. Since a single-stage OTA is a single-pole system, A_o , ω_{3dB} , and ω_{GB} are enough to determine the frequency response of a single-stage OTA. Moreover, ω_{3dB} and ω_{GB} as well as A_o are obtained from the G_m and r_{out} . A_o , ω_{3dB} , and ω_{GB} of a five-transistor OTA, represented as $A_{o,5TR_OTA}$, $\omega_{3dB,5TR_OTA}$, and $\omega_{GB,5TR_OTA}$, are expressed as follows:

$$A_{o,5TR_OTA} = -G_{m,5TR_OTA} \cdot r_{out,5TR_OTA} = -\frac{1}{2} \cdot g_m \cdot r_o, \quad (3)$$

$$\omega_{3dB,5TR_OTA} = \frac{1}{r_{out,5TR_OTA} \cdot C_{Load}} = \frac{1}{(r_o/2) \cdot C_{Load}}, \quad (4)$$

$$\omega_{GB,5TR_OTA} = |A_{o,5TR_OTA}| \cdot \omega_{3dB,5TR_OTA} = \frac{g_m}{C_{Load}}, \quad (5)$$

where C_{Load} is the capacitance of the load capacitor. At DC, the load capacitor is opened, generating a voltage from $G_{m,5TR_OTA}$ and $r_{out,5TR_OTA}$. As the frequency increases, the load capacitor cannot be ignored, forming a pole with $r_{out,5TR_OTA}$.

The next performance metrics, the input and output swing ranges, are illustrated in Figure 2c. Here, $V_{D,SAT}$ is the minimum V_{ds} to maintain the saturation operation of the transistors. And V_{gs} is set to $V_{D,SAT} + V_{th}$ at a minimum, where V_{th} is the threshold voltage of the transistor. When $V_{D,SAT}$ and V_{th} of all transistors are same, the input voltage signal must be higher than $2V_{D,SAT} + V_{th}$ and lower than $V_{DD} - V_{D,SAT}$. In the same condition, since two NMOSs (M_2 and M_5) are operated in a saturation region, the lowest output voltage level is $2V_{D,SAT}$. The highest output voltage level is limited to $V_{DD} - V_{D,SAT}$ since M_4 should be operated in the saturation region. Compared to other types of single-stage OTAs, a five-transistor OTA has moderate input and output swing ranges that are neither wide nor narrow. Especially when the input DC bias voltage is relatively low, a PMOS-based five-transistor OTA is more suitable.

To look into other important performance metrics, SR_R and SR_F , unity-gain buffers are designed by five-transistor OTAs as shown in Figure 2d,e. In Figure 2d, to derive the SR_R of a five-transistor OTA, represented as $SR_{R,5TR_OTA}$, a step response is applied with a very short rising time. To derive the falling SR_F of a five-transistor OTA, represented as $SR_{F,5TR_OTA}$, a step response is applied with a very short falling time, as shown in Figure 2e. This means that the signal is much faster than the OTA bandwidth (see Figure 2d first). When a step response is applied with a rising edge, M_1 turns on and M_2 turns off. Then, all the current flowing through M_5 flows through M_1 , which is copied to M_4 by the current source that is composed of M_3 and M_4 . Since the current transferred to M_4 does not flow to M_2 but flows to C_{Load} , the output voltage rises linearly along with the step response. Conversely, when a step response is applied with a falling edge, M_1 turns off and M_2 turns on. I_{D5} does not flow to M_1 , M_3 , and M_4 , but flows to C_{Load} through M_2 . The output voltage decreases linearly along with the step response. Note that the output voltage is a result of current accumulation on the capacitor. As a result, $SR_{R,5TR_OTA}$ and $SR_{F,5TR_OTA}$ are expressed as follows:

$$SR_{R,5TR_OTA} = SR_{F,5TR_OTA} = \frac{I_{D5}}{C_{Load}}. \quad (6)$$

The next performance metrics that we analyze are nonidealities including circuit noise, offset, and mismatches. Since the voltage-to-current conversion and amplification are mainly performed by M_1 and M_2 , the nonidealities of these two input transistors become the most dominant sources that cause performance degradation. In contrast, the nonidealities of M_5 become less dominant to the overall performance since they can be canceled out at the output port with perfectly matched M_3 and M_4 . As a result, the effect of the nonidealities of M_1 and M_2 on the overall performance is greater than that of M_3 and M_4 , and that of M_3 and M_4 is greater than that of M_5 . Therefore, in general, circuit designers reduce the performance degradation from the nonidealities by increasing the size of M_1 and M_2 . Moreover, since the mismatch between M_3 and M_4 cause a systematic offset, a careful layout is needed to reduce the mismatch.

3.2. Telescopic Cascode OTA

The telescopic cascode OTA in Figure 3 is a type of single-stage OTA that achieves a higher r_{out} and A_o compared to those of a five-transistor OTA. In particular, the telescopic cascode OTA can obtain a high r_{out} and A_o without an additional stage, thereby achieving power efficiency as well.

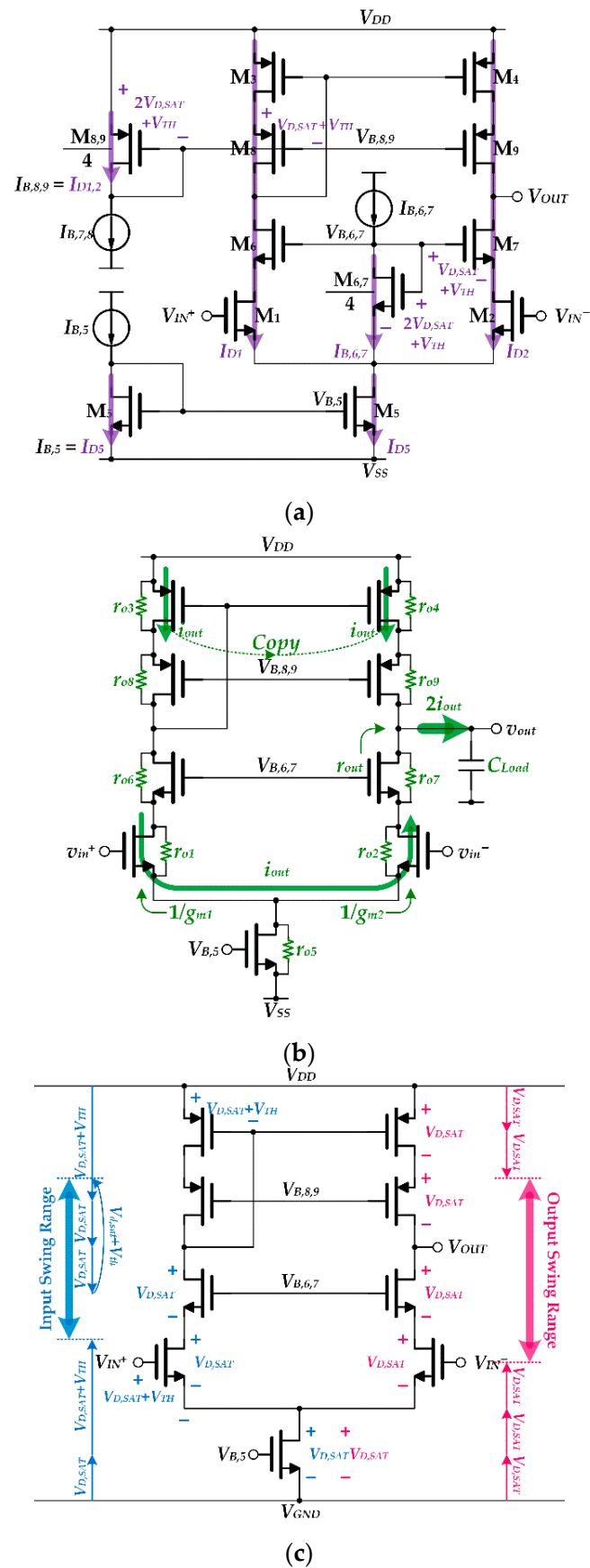


Figure 3. Cont.

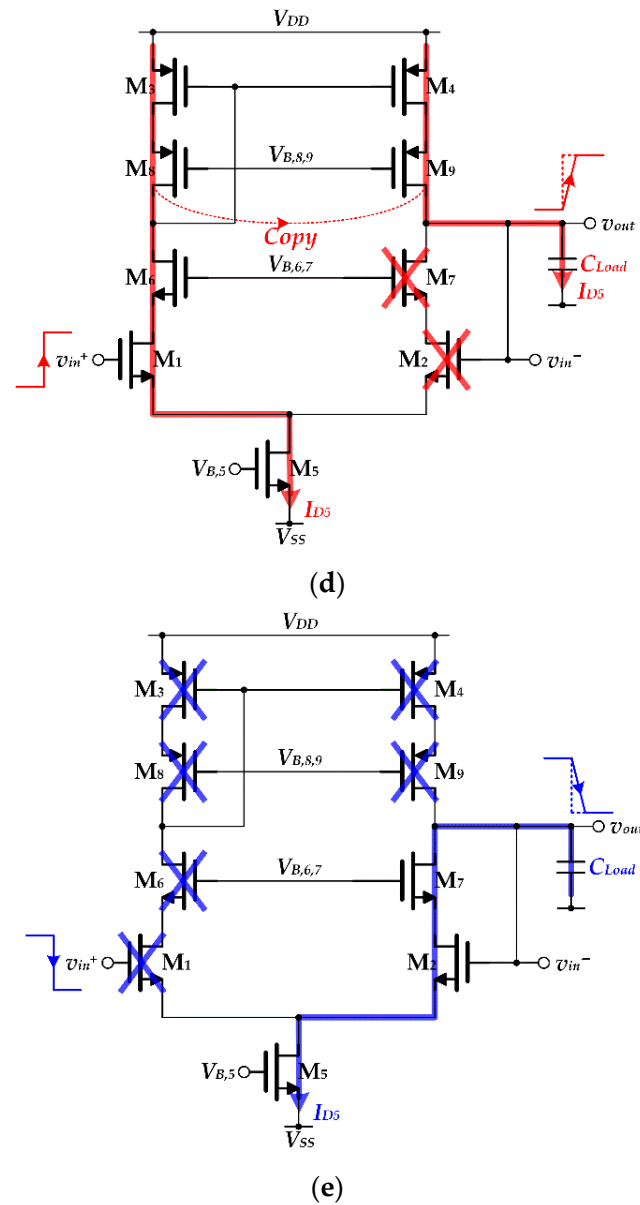


Figure 3. Telescopic cascode OTA for showing (a) operation at equilibrium, (b) small-signal parameters and frequency response, (c) input and output swing ranges, (d) rising SR, and (e) falling SR.

Like in a five-transistor OTA, M_1 and M_2 generate current signals, represented as I_{D1} and I_{D2} , respectively. And in a cascode current mirror composed of M_3 , M_4 , M_8 , and M_9 , I_{D1} is copied to M_4 . As shown in Figure 3a, when two input signals have the same voltage level, no current flows to the load. In contrast, when v_{id} is applied to the input nodes, a current flows to the load due to the different directions of I_{D1} and I_{D2} .

See Figure 3b for a small signal analysis with a differential input signal. The principle of the voltage-to-current conversion of a telescopic cascode OTA is the same as that of a five-transistor OTA. Therefore, when M_1 and M_2 are matched with g_m , the G_m of a telescopic cascode OTA, G_{m,TC_OTA} , is expressed as follows:

$$G_{m,TC_OTA} = g_{m1} = g_{m2} = g_m. \quad (7)$$

In contrast, the r_{out} and A_o are boosted further thanks to the cascode current mirror. At the output node, the r_{out} formed by two NMOSs and the r_{out} formed by two PMOSs are

connected in parallel. Therefore, when transistors are matched with the same g_m and r_o , the r_{out} of a telescopic cascode OTA, r_{out,TC_OTA} , is expressed as follows:

$$r_{out,TC_OTA} = (r_{o4} \cdot g_{m9} \cdot r_{o9}) \parallel \left\{ \left(\frac{1}{g_{m1}} \cdot g_{m2} \cdot r_{o2} \right) \cdot g_{m7} \cdot r_{o7} \right\} = \frac{1}{2} \cdot g_m \cdot r_o^2. \quad (8)$$

The r_{out,TC_OTA} is $g_m r_o$ times higher than $r_{out,5TR_OTA}$ in (2). The r_{out} is amplified by the voltage gain of a single transistor through the cascode current mirror.

Since a telescopic cascode OTA is also a type of single-pole system, A_o , ω_{3dB} , and ω_{GB} determine the frequency response. A_o , ω_{3dB} , and ω_{GB} of a telescopic cascode OTA, represented as A_{o,TC_OTA} , ω_{3dB,TC_OTA} , and ω_{GB,TC_OTA} , are expressed as follows:

$$A_{o,TC_OTA} = -G_m \cdot r_{out} = -\frac{1}{2} \cdot (g_m \cdot r_o)^2, \quad (9)$$

$$\omega_{3dB,TC_OTA} = \frac{1}{r_{out,TC_OTA} \cdot C_{Load}} = \frac{1}{(g_m \cdot r_o^2 / 2) \cdot C_{Load}}, \quad (10)$$

$$\omega_{GB,TC_OTA} = |A_{o,TC_OTA}| \cdot \omega_{3dB,TC_OTA} = \frac{g_m}{C_{Load}}. \quad (11)$$

A_{o,TC_OTA} in (9) is $g_m r_o$ times higher than $A_{o,5TR_OTA}$ in (3), while ω_{3dB,TC_OTA} in (10) is $g_m r_o$ times lower than $\omega_{3dB,5TR_OTA}$ in (4). Note that ω_{GB,TC_OTA} is same as $\omega_{GB,5TR_OTA}$ in (5). The gain of a telescopic cascode OTA is higher than that of a five-transistor OTA for the frequency range less than $\omega_{3dB,5TR_OTA} / 2\pi$. In contrast, the gain of a telescopic cascode OTA is the same as that of a five-transistor OTA for the frequency range from $\omega_{3dB,5TR_OTA} / 2\pi$ to ω_{GB,TC_OTA} .

Although the cascode current mirror improves the gain and r_{out} further, it degrades the input and output swing ranges. Among various types of bias circuits, the bias circuits shown in Figure 3a are employed to minimize the degradation of the swing range (see M_8 and M_9 first). M_8 and M_9 are biased by a diode-connected PMOS with a current source, where the transistor size of the diode-connected PMOS is four times smaller than those of M_5 and M_9 . This diode-connected PMOS generates a voltage difference of $2V_{D,SAT} + V_{th}$ from V_{DD} , which is the minimum voltage that allows for both transistors to operate in the saturation region. In this bias condition, M_8 and M_9 can be operated with minimum $|V_{ds}|$ of $V_{D,SAT}$. Similarly, to allow for M_2 and M_7 to operate in the saturation region with a minimum $|V_{ds}|$ of $V_{D,SAT}$, a diode-connected NMOS with a current source where the transistor size of the diode-connected NMOS is four times smaller than those of M_6 and M_7 is employed. When the $V_{D,SAT}$ and V_{th} of all transistors are the same, the input voltage signal must be higher than $2V_{D,SAT} + V_{th}$ and lower than $V_{DD} - 2V_{D,SAT}$. In the same condition, since three NMOSs are cascoded at the output, the lowest output voltage level is $3V_{D,SAT}$. Similarly, the highest output voltage level is limited to $V_{DD} - 2V_{D,SAT}$ due to two cascoded PMOSs at the output node. Compared to the swing ranges of a five-transistor OTA, the telescopic cascode OTA has a more narrow swing range due to the cascade current mirror. Moreover, as shown in Figure 3a, the complexity and power consumption of bias circuits are increased. That is, the gain and r_{out} of a telescopic cascode OTA are improved at the sacrifice of swing ranges and the complexity of bias circuits.

The SR_R and SR_F of a telescopic cascode OTA are analyzed in the same way as a five-transistor OTA. As shown in Figure 3d,e, in the unity-gain buffer configuration, the additional cascode transistors do not affect the SR_R and SR_F . M_6 , M_7 , M_8 , and M_9 just follow the operation of M_1 , M_2 , M_3 , and M_4 , respectively. The SR_R and SR_F of a telescopic cascode OTA, represented as SR_{R,TC_OTA} and SR_{F,TC_OTA} , are expressed as follows:

$$SR_{R,TC_OTA} = SR_{F,TC_OTA} = \frac{I_{D5}}{C_{Load}}. \quad (12)$$

The nonidealities of additional transistors, M_6 – M_9 , cause negligible performance degradation. The nonidealities of additional transistors, M_6 – M_9 , are transferred to the output with a low gain. Therefore, as in a five-transistor OTA, the effect of the nonidealities of M_1 and M_2 on the overall performance is greater than that of M_3 and M_4 , and that of M_3 and M_4 is greater than that of M_5 .

3.3. Folded Cascode OTA

The folded cascode OTA shown in Figure 4 is employed to extend the input swing ranges of the telescopic cascode OTA while maintaining a high gain and high r_{out} . The transformation from a telescopic cascode OTA to a folded cascode OTA involves modifying the transistor configuration and biasing arrangement. In the NMOS-input telescopic cascode OTA, the input stage consists of a differential pair, followed by NMOS cascode transistors connected to the load, as shown in Figure 3. On the other hand, the folded cascode OTA replaces the NMOS cascode transistors with four PMOS transistors configured in a folded arrangement between the input stage and the output stage. Two PMOS transistors, M_8 and M_9 , serve as the cascode transistors, while the other PMOS transistors, M_3 and M_4 , act as the current sources. The current sources provide bias currents to the input stage and output stage.

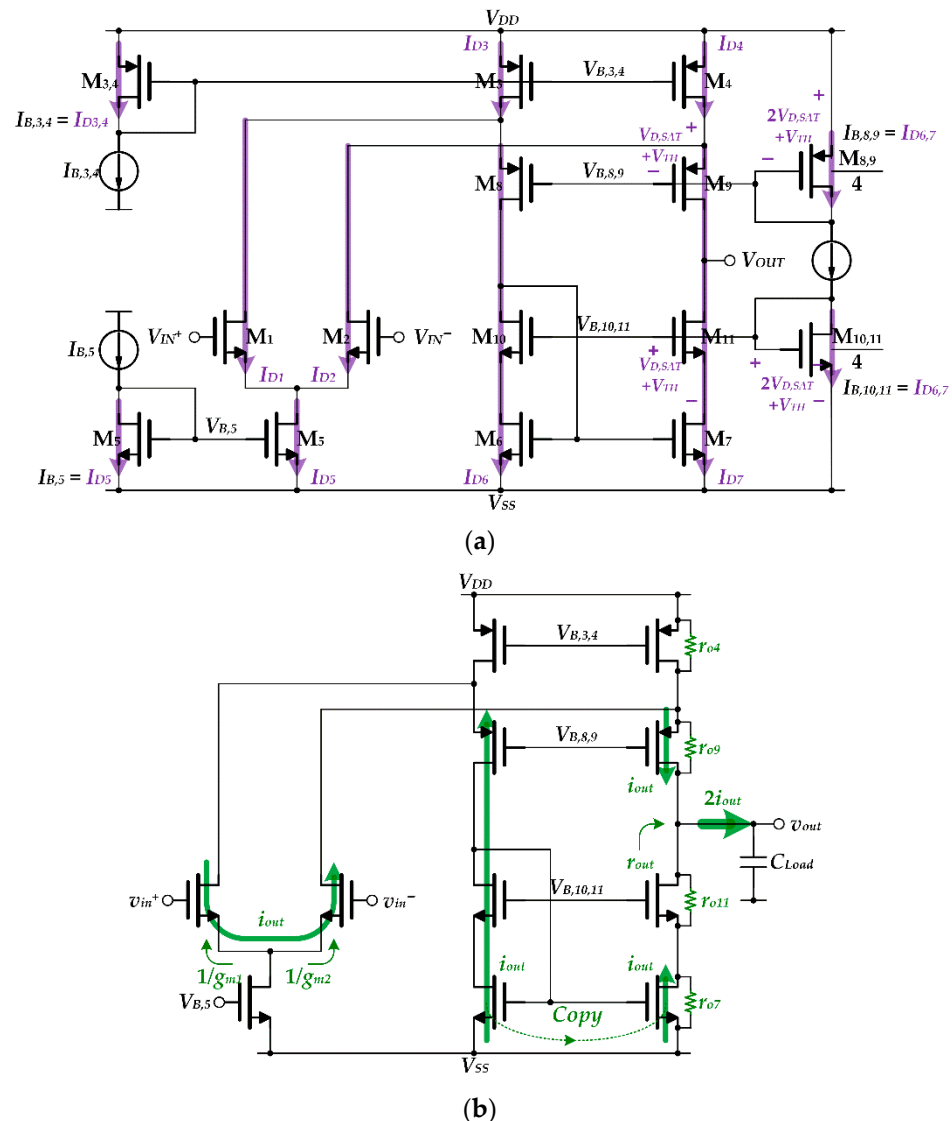


Figure 4. Cont.

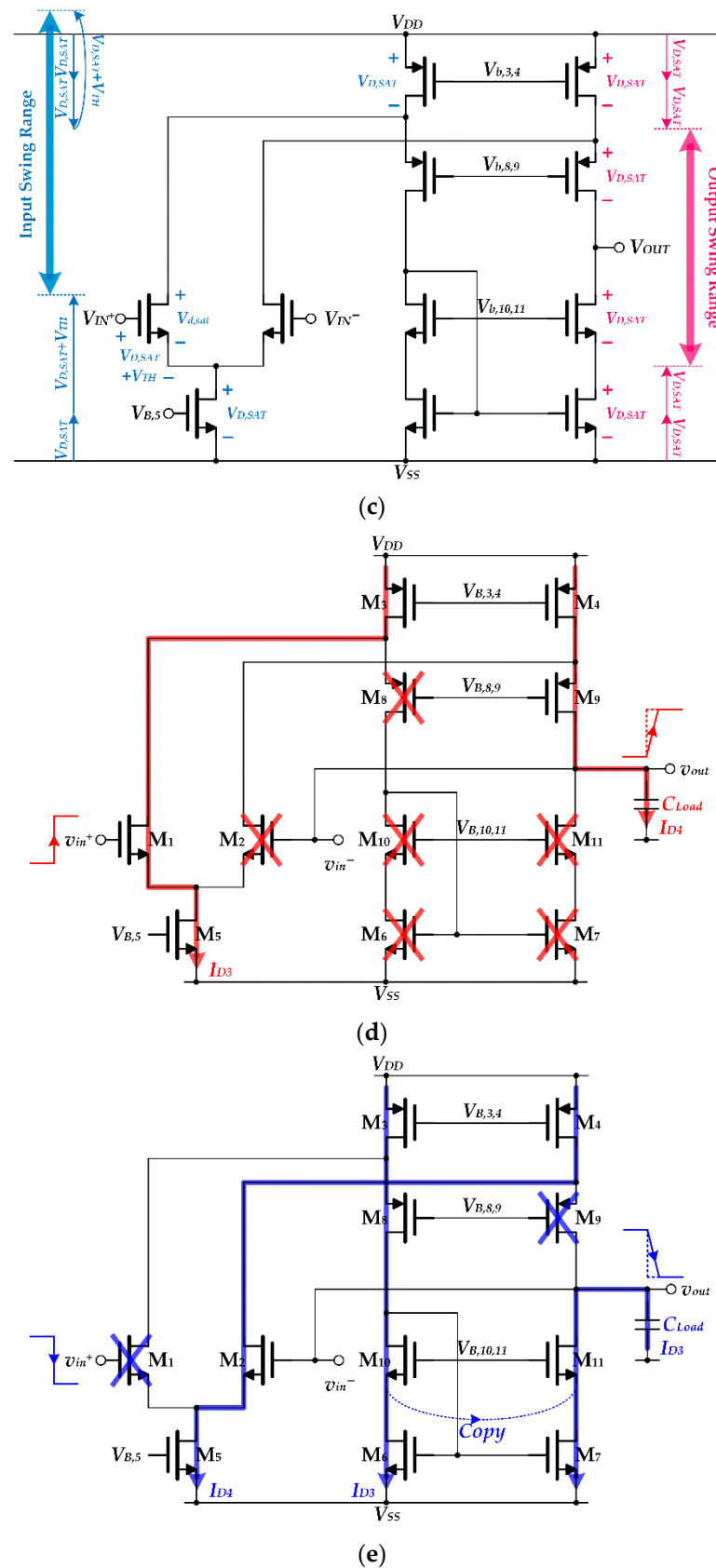


Figure 4. Folded cascode OTA for showing (a) operation at equilibrium, (b) small-signal parameters and frequency response, (c) input and output swing ranges, (d) rising SR, and (e) falling SR. In (d,e), $I_{D3} = I_{D4} = I_{D5}$.

The modification improves the input swing range, but as shown in Figures 3a and 4a, one more diode-connected transistor is required when the same wide-swing cascode current mirrors are employed. Fortunately, the bias circuit for M_8 and M_9 and the bias circuit for M_{10} and M_{11} can share one bias current, consuming the same power compared to the bias circuits for a telescopic cascode OTA.

See Figure 4a first to show the operation at equilibrium. The current flowing through M_9 , denoted as I_{D9} , is the same as $I_{D4} - I_{D2}$, according to Kirchhoff's Current Law (KCL). The current flowing through M_6 , denoted as I_{D6} , is copied to M_7 , where I_{D6} is $I_{D3} - I_{D1} = I_{D4} - I_{D2}$. Therefore, like a five-transistor OTA and a telescopic cascode OTA, no current flows to the load when two input signals have the same voltage level.

The small-signal parameters are also the same as those of a telescopic cascode OTA. See Figure 4b for a small signal analysis with differential input signals. The principle of a voltage-to-current conversion of a folded cascode OTA is the same as that of a telescopic cascode OTA. Since the differential voltage inputs are converted to a current by M_1 and M_2 , the G_m of a folded cascode OTA, G_{m,FC_OTA} , is expressed as follows:

$$G_{m,FC_OTA} = g_{m1} = g_{m2} = g_m \quad (13)$$

where M_1 and M_2 are matched with g_m . Likewise, the r_{out} is boosted further thanks to the cascode current mirrors. At the output node, the r_{out} formed by M_7 and M_{11} and the r_{out} formed by two M_2 and M_9 are connected in parallel. Since M_9 works as a common-gate amplifier from the input signal point of view like M_{11} , M_2 and M_9 also increase the r_{out} . Therefore, when the transistors are matched with the same g_m and r_o , the r_{out} of a folded cascode OTA, r_{out,TC_OTA} , is expressed as follows:

$$r_{out,FC_OTA} = (r_{o4} \cdot g_{m9} \cdot r_{o9}) || (r_{o7} \cdot g_{m11} \cdot r_{o11}) = \frac{1}{2} \cdot g_m \cdot r_o^2. \quad (14)$$

Since the small-signal parameters are same as those of a telescopic cascode OTA, A_o , ω_{3dB} , and ω_{GB} of a folded cascode OTA, denoted as A_{o,TC_OTA} , ω_{3dB,TC_OTA} , and ω_{GB,TC_OTA} , are expressed as follows:

$$A_{o,FC_OTA} = -G_m \cdot r_{out} = -\frac{1}{2} \cdot (g_m \cdot r_o)^2, \quad (15)$$

$$\omega_{3dB,FC_OTA} = \frac{1}{r_{out,FC_OTA} \cdot C_{Load}} = \frac{1}{(g_m \cdot r_o^2 / 2) \cdot C_{Load}}, \quad (16)$$

$$\omega_{GB,FC_OTA} = |A_{o,FC_OTA}| \cdot \omega_{3dB,FC_OTA} = \frac{g_m}{C_{Load}}. \quad (17)$$

The same frequency response characteristic as that of a telescopic cascode structure is expected.

As shown in Figure 4c, the folded cascode OTA has a wider input swing range compared to the telescopic cascode OTA due to the use of a PMOS cascode transistor in the folded arrangement. In a telescopic cascode OTA, the NMOS cascode transistor, M_9 , restricts the input swing range as it requires a minimum voltage headroom. However, the PMOS cascode transistor in a folded cascode OTA, M_9 , allows for the input voltage to swing more positively, expanding the input range and enabling operation in a wider voltage range. In contrast, the output swing range of a folded cascode OTA is same as that of a telescopic cascode OTA since M_4 , M_9 , M_7 , and M_{11} require a minimum voltage headroom.

Figure 4d,e show the slewing of a folded cascode OTA for the rising and falling step responses, respectively. In general, I_{D3} and I_{D4} are set to be higher than I_{D5} [21]. Let us first look at Figure 4d,e, where I_{D3} and I_{D4} are set to be same as I_{D5} . The rising step response turns on M_1 , allowing for M_1 and M_5 to become sinking paths of I_{D3} . No current flows to M_8 due to I_{D3} and I_{D5} being equal, which turns off M_2 , and the NMOS cascode current mirror is composed of M_6 , M_7 , M_{10} , and M_{11} . As a result, according to the KCL, the

I_{D4} flows to C_{Load} through M_9 , and the rising SR of a folded cascode OTA, SR_{R,FC_OTA} , is expressed as follows:

$$SR_{R,FC_OTA} = \frac{I_{D4}}{C_{Load}} = \frac{I_{D5}}{C_{Load}}. \quad (18)$$

When the falling step response is applied, M_1 turns off and M_2 turns on, allowing for M_2 and M_5 to become sinking paths of I_{D4} and turning off M_9 . Therefore, the NMOS cascode current mirror composed of M_6 , M_7 , M_{10} , and M_{11} sinks the current from C_{Load} , where the sinking current magnitude is I_{D3} . As a result, the falling SR of a folded cascode OTA, SR_{F,FC_OTA} , is expressed as follows:

$$SR_{F,FC_OTA} = \frac{I_{D3}}{C_{Load}} = \frac{I_{D5}}{C_{Load}}. \quad (19)$$

where $I_{D3} = I_{D4} = I_{D5}$, and SR_{R,FC_OTA} and SR_{F,FC_OTA} are the same as SR_{R,FC_OTA} and SR_{F,FC_OTA} , respectively.

When $I_{D3} = I_{D4} > I_{D5}$ and the rising step response is applied, M_2 does not turn off perfectly, flowing to $I_{D3} - I_{D5}$. Therefore, SR_{R,FC_OTA} also follows (18) despite the increasing current. When $I_{D3} = I_{D4} > I_{D5}$ and the rising step response is applied, the sourcing current $I_{D4} - I_{D5}$ flows through M_9 , which has a different direction of the sinking current I_{D3} flowing through M_7 and M_{11} . SR_{F,FC_OTA} also follows (19) despite the increasing current. As a result, setting I_{D3} and I_{D4} as larger than I_{D5} to increase SR_{R,FC_OTA} and SR_{F,FC_OTA} is not effective since SR_{R,FC_OTA} and SR_{F,FC_OTA} are independent of I_{D3} and I_{D4} .

The performance degradation due to the nonidealities of a folded cascode OTA includes the performance degradation due to telescopic nonidealities. Like the telescopic cascode amplifier, the nonidealities of cascoded transistors, M_8 – M_{11} , cause negligible performance degradation due to the low gain. As in a folded cascode OTA, the effect caused by the nonidealities of M_1 and M_2 on the overall performance is greater than those caused by other transistors. However, in general, the performance degradation by two additional current sources, M_3 and M_4 , are not negligible, thus reducing the nonidealities of M_3 and M_4 by increasing the gate size, and so on. This added noise is one of the disadvantages of a folded cascode OTA.

3.4. Current Mirror OTA

Telescopic and folded cascode OTAs are employed to increase the r_{out} , thus achieving a high A_o . However, as a trade-off, the output swing ranges of these two OTAs and one of the performance metrics related to speed, ω_{GB} , are limited due to the cascode output stages. Furthermore, another performance metric related to speed, SR, is also limited by the magnitude of the input bias current.

See Figure 5a to look into the design of the bias circuit and operation at equilibrium. Like a five-transistor OTA, the current source, $I_{b,5}$, is copied to M_5 via the diode-connected transistor. Since M_3 and M_4 are diode connected, the V_{GS} of M_3 and M_4 are determined by I_5 and the aspect ratios of M_3 and M_4 . Through the current mirrors, the V_{GS} of other transistors are also determined without additional bias circuits. Like a five-transistor OTA, this simple bias circuit becomes one of the advantages of a five-transistor OTA compared to OTAs based on a cascode structure. At the equilibrium, M_1 and M_2 generate current signals, I_{D1} and I_{D2} , respectively. I_{D1} is fed into M_3 , and I_{D6} flowing through M_6 becomes $K \cdot I_{D1}$ via the current mirror composed of M_3 and another transistor, which is K times wider than M_3 . When M_6 and M_7 have the same aspect ratios, I_{D7} flowing through M_7 also becomes $K \cdot I_{D1}$. Note that I_{D2} is equal to I_{D1} . Since I_{D2} is amplified by the current mirror composed of M_4 and another transistor, which is K times wider than M_4 , no current flows to the load. In contrast, when two input signals have the same DC voltage level with a small AC differential signal, denoted as v_{id} , a current flows to the load due to the different directions of I_{D1} and I_{D2} .

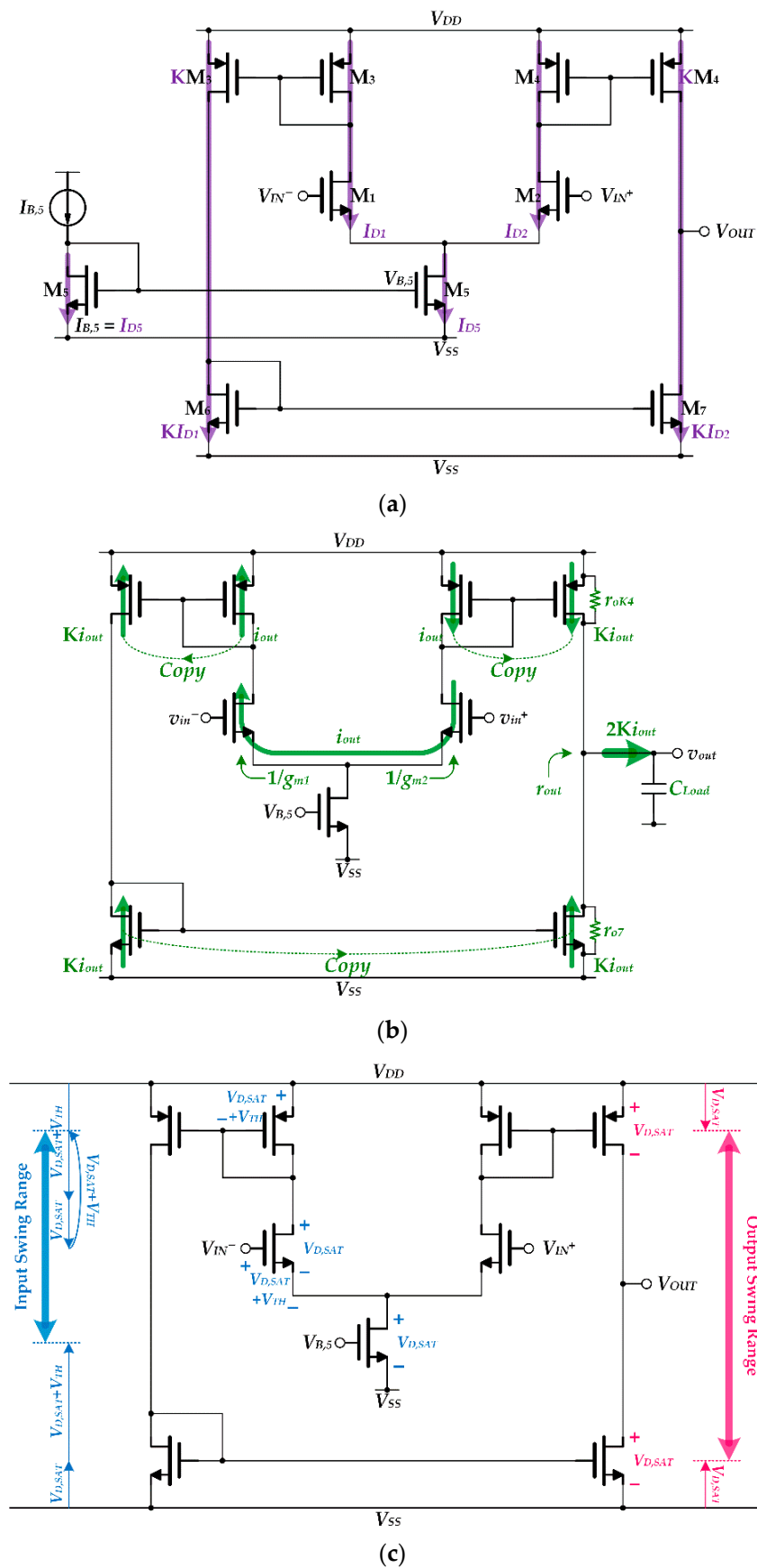


Figure 5. Cont.

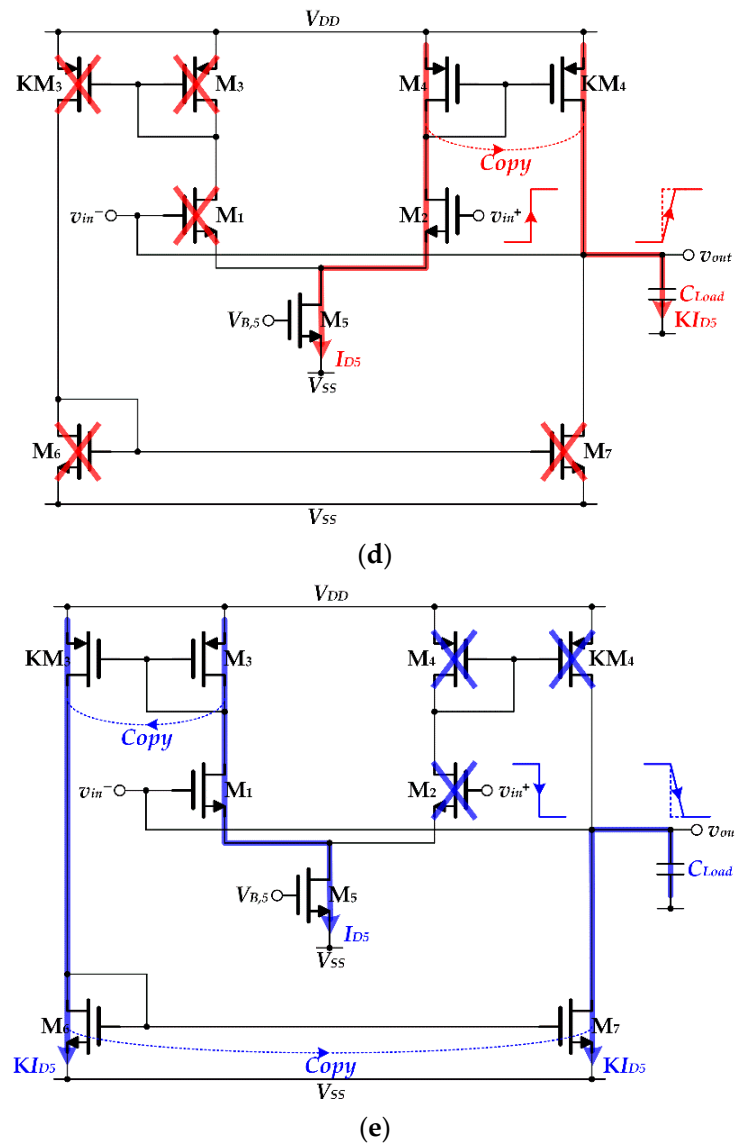


Figure 5. Current mirror OTA for showing (a) operation at equilibrium, (b) small-signal parameters and frequency response, (c) input and output swing ranges, (d) rising SR, (e) and falling SR.

See Figure 4b for a small signal analysis with a differential input signal. When M_1 and M_2 are matched with the same transconductance, denoted as g_m , M_1 and M_2 generate $i_{out} = g_m(v_{id}/2)$ and $-i_{out}$. i_{out} and $-i_{out}$ are amplified to $K \cdot i_{out}$ and $-K \cdot i_{out}$ at the output node. Two amplified currents are merged together at the output node, thereby flowing $2i_{out} = K \cdot g_m v_{id}$ to the load. In this regard, the G_m of a current mirror OTA, G_{m,CM_OTA} , is expressed as follows:

$$G_{m,CM_OTA} = K \cdot g_{m1} = K \cdot g_{m2} = K \cdot g_m. \quad (20)$$

Thanks to the current mirrors at the output stage, a higher G_m can be achieved compared to other OTAs from the same g_{m1} and g_{m2} . However, at the expense of a higher G_m , additional current paths for amplifying the current are required, consuming higher power.

Let us look into the r_{out} of a current mirror OTA. At the output node, r_{oK4} and r_{o7} are connected in parallel, where r_{oK4} is the output impedance of the transistor, which is K times wider than M_4 . Therefore, two transistors at the output node have the same output

impedance, denoted as r_o , and the r_{out} of a five-transistor OTA, denoted as r_{out,CM_OTA} , is expressed as follows:

$$r_{out,CM_OTA} = r_{oK4} || r_{o7} = \frac{1}{2} \cdot r_o. \quad (21)$$

The frequency response of a current mirror OTA can be derived from small-signal parameters. A_o , ω_{3dB} , and ω_{GB} of a current mirror OTA, denoted as A_{o,CM_OTA} , ω_{3dB,CM_OTA} , and ω_{GB,CM_OTA} , are expressed as follows:

$$A_{o,CM_OTA} = -G_{m,CM_OTA} \cdot r_{out,CM_OTA} = -\frac{K}{2} \cdot g_m \cdot r_o, \quad (22)$$

$$\omega_{3dB,CM_OTA} = \frac{1}{r_{out,CM_OTA} \cdot C_{Load}} = \frac{1}{(r_o/2) \cdot C_{Load}}, \quad (23)$$

$$\omega_{GB,CM_OTA} = |A_{o,CM_OTA}| \cdot \omega_{3dB,CM_OTA} = \frac{K \cdot g_m}{C_{Load}}. \quad (24)$$

Here, r_o in (21)–(23) is K times lower than r_o in (2)–(4) since r_o of a single transistor is inversely proportional to the bias current at equilibrium. In this regard, r_{out,CM_OTA} in (21) is K times lower than $r_{out,5TR_OTA}$ in (21). As a result, theoretically, A_{o,CM_OTA} in (22) is the same as $A_{o,5TR_OTA}$ in (3). In contrast, ω_{3dB,CM_OTA} in (23) and ω_{GB,CM_OTA} in (24) are K times lower than $\omega_{3dB,5TR_OTA}$ in (4) and $\omega_{GB,5TR_OTA}$ in (5), respectively.

A current mirror OTA has a wider output swing range compared to other types of OTAs, which becomes one of the advantages of a current mirror OTA. There are only two transistors at the output node, thus guaranteeing that those two transistors operate in the saturation region. The highest output voltage level is limited to $V_{DD} - V_{D,SAT}$, and the lowest output voltage level is limited to $V_{D,SAT}$. Since other single-stage OTAs have a larger number of transistors from 3 to 5, a current mirror OTA has the widest output swing ranges among single-stage OTAs. Especially when the input DC bias voltage is relatively low, a PMOS-based five-transistor OTA is more suitable. It can be said that a current mirror OTA can guarantee the rail-to-rail output swing range. The input swing range of a current mirror OTA is the same as that of a five-transistor OTA since these two OTAs have a similar structure at the input node.

The current amplification by K times also enhances the SR. As shown in Figure 5d,e, the current flowing through PMOS at the output node determines a rising SR in a current mirror OTA, denoted as SR_{R,CM_OTA} , and the current flowing through the NMOS at the output node determines a rising SR in a current mirror OTA, denoted as SR_{F,CM_OTA} . Since I_{D5} is amplified by K times through current mirrors, SR_{R,CM_OTA} and SR_{F,CM_OTA} are expressed as follows:

$$SR_{R,CM_OTA} = SR_{F,CM_OTA} = K \cdot \frac{I_{D5}}{C_{Load}}. \quad (25)$$

while the SRs of other OTAs are limited by the magnitude of the input bias current, the SR of a current mirror OTA is not limited by the magnitude of the input bias current. However, the higher the K , the higher the power consumption.

Despite the advantages of a current mirror OTA including a larger G_m , a wider bandwidth, a wider output swing range, and a faster SR, a current mirror OTA suffers greater performance degradation due to nonidealities compared to other structures. Nonidealities of all additional current mirrors are amplified to the output with a moderate gain. Therefore, a current mirror OTA has not been employed as a low-noise amplifier in applications that require a high resolution.

3.5. Summary

Table 1 summarizes the performance of single-stage OTAs. As mentioned earlier, the r_o in a current mirror OTA is K times lower than the r_o in other transistors since the r_o of a single transistor is inversely proportional to the bias current at equilibrium.

Table 1. Summary of theoretical performance metrics.

Topology	5-Transistor OTA	Telescopic Cascode OTA	Folded Cascode OTA	Current Mirror OTA
G_m	g_m	g_m	g_m	$K \cdot g_m$
r_{out}	$\frac{1}{2} \cdot r_o$	$\frac{1}{2} \cdot g_m \cdot r_o^2$	$\frac{1}{2} \cdot g_m \cdot r_o^2$	$\frac{1}{2} \cdot r_o$
A_o	$-\frac{1}{2} \cdot g_m \cdot r_o$	$-\frac{1}{2} \cdot (g_m \cdot r_o)^2$	$-\frac{1}{2} \cdot (g_m \cdot r_o)^2$	$-\frac{K}{2} \cdot g_m \cdot r_o$
ω_{3dB}	$\frac{1}{(r_o/2) \cdot C_{Load}}$	$\frac{1}{(g_m \cdot r_o^2/2) \cdot C_{Load}}$	$\frac{1}{(g_m \cdot r_o^2/2) \cdot C_{Load}}$	$\frac{1}{(r_o/2) \cdot C_{Load}}$
ω_{GB}	$\frac{g_m}{C_{Load}}$	$\frac{g_m}{C_{Load}}$	$\frac{g_m}{C_{Load}}$	$K \cdot \frac{g_m}{C_{Load}}$
SR_R	$\frac{I_{D5}}{C_{Load}}$	$\frac{I_{D5}}{C_{Load}}$	$\frac{I_{D5}}{C_{Load}}$	$K \cdot \frac{I_{D5}}{C_{Load}}$
SR_F	$\frac{I_{D5}}{C_{Load}}$	$\frac{I_{D5}}{C_{Load}}$	$\frac{I_{D5}}{C_{Load}}$	$K \cdot \frac{I_{D5}}{C_{Load}}$

A five-transistor OTA has the simplest architecture among the four types of single-stage OTAs. Compared to a five-transistor OTA, telescopic cascode and folded cascode OTAs achieve a larger r_{out} , which results in a higher A_o . A current mirror OTA achieves higher ω_{3dB} , ω_{GB} , SR_R , and SR_F compared to other types of OTAs. That is, most of the performance metrics related to the speed of OTAs are improved thanks to the current amplification via current mirrors. Moreover, a higher G_m is also achieved. However, these improvements come at the expense of power consumption. Another advantage of a current mirror OTA is the wide output swing range. Particularly, regardless of the power consumption, it can obtain the widest output swing range thanks to a small number of transistors.

4. Results

Four types of single-stage OTAs, which were analyzed in Section 3, were designed and simulated in a 180 nm complementary metal–oxide–semiconductor (CMOS) process with a V_{DD} of 1.8 V to verify the theoretical analysis. For a fair performance comparison, the input transistors of OTAs, M_1 and M_2 , have the same g_m with the same bias current of 1 μ A. Table 2 shows the transistor sizes of the designed single-stage OTAs. $V_{D,SAT}$ is set to about 100 mV so that all transistors operate in the strong inversion region.

Table 2. Transistor sizes of designed single-stage OTAs.

	5-Transistor OTA	Telescopic Cascode OTA	Folded Cascode OTA	Current Mirror OTA
$M_1(W/L)$	1 μ /1 μ	1 μ /1 μ	1 μ /1 μ	1 μ /1 μ
$M_2(W/L)$	1 μ /1 μ	1 μ /1 μ	1 μ /1 μ	1 μ /1 μ
$M_3(W/L)$	4 μ /1 μ	4 μ /1 μ	8 μ /1 μ	4 μ /1 μ
$M_4(W/L)$	4 μ /1 μ	4 μ /1 μ	8 μ /1 μ	4 μ /1 μ
$M_5(W/L)$	2 μ /1 μ	3 μ /1 μ	2 μ /1 μ	2 μ /1 μ
$M_6(W/L)$	N/A	1 μ /1 μ	1 μ /1 μ	3 μ /1 μ
$M_7(W/L)$	N/A	1 μ /1 μ	1 μ /1 μ	3 μ /1 μ
$M_8(W/L)$	N/A	4 μ /1 μ	4 μ /1 μ	N/A
$M_9(W/L)$	N/A	4 μ /1 μ	4 μ /1 μ	N/A
$M_{10}(W/L)$	N/A	N/A	1 μ /1 μ	N/A
$M_{11}(W/L)$	N/A	N/A	1 μ /1 μ	N/A

Figure 6 shows the simulation results of a designed five-transistor OTA. Figure 6a shows the frequency response, and Figure 6b shows the rising slewing of an OTA with

SR_R when C_{Load} is set to 1 pF. Figure 6c shows an input-referred noise, which is one of the representative nonidealities of an OTA. Figure 7, Figure 8, and Figure 9 show the same performance metrics of a telescopic cascode OTA, a folded cascode OTA, and a current mirror OTA, respectively.

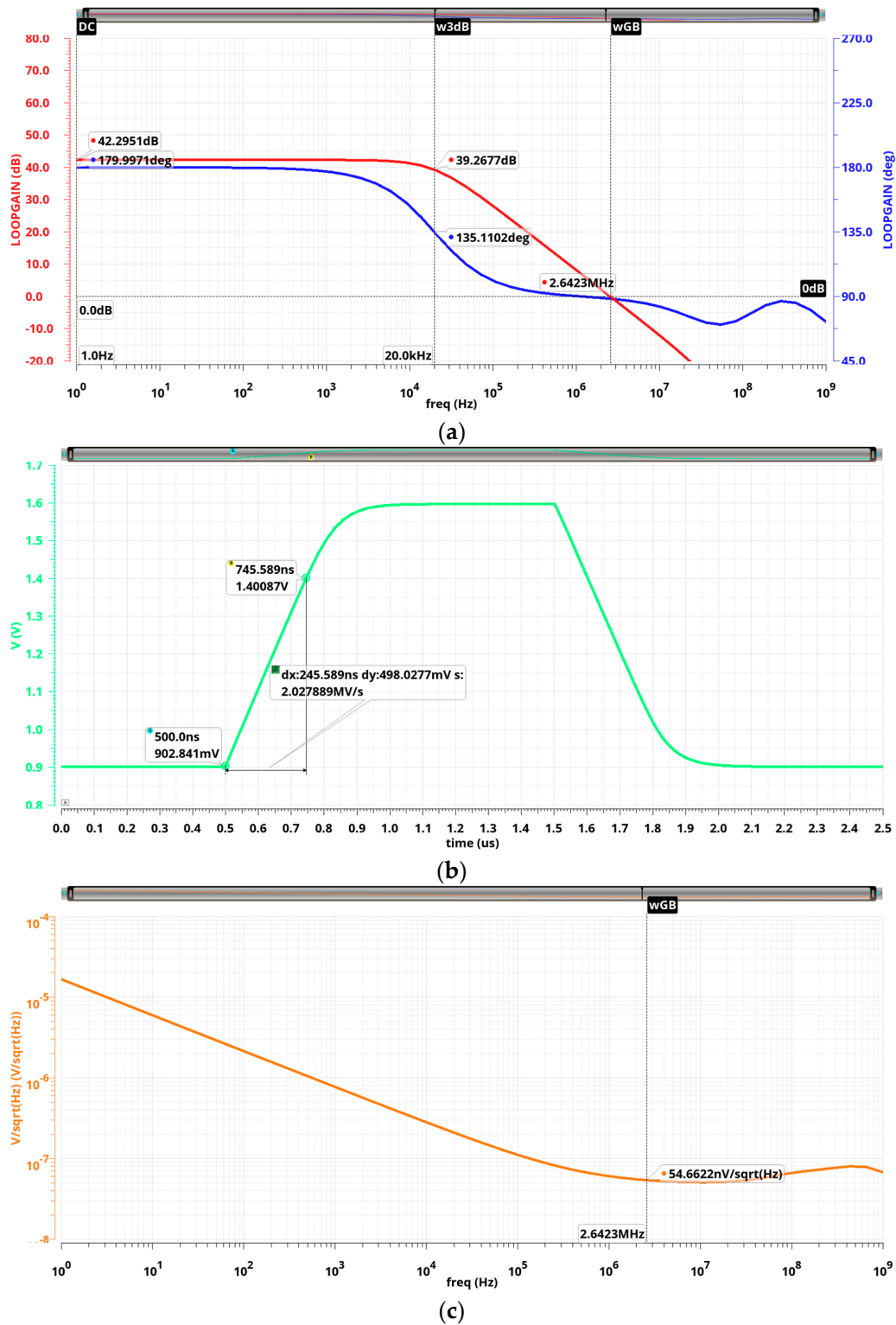


Figure 6. Simulation results of 5-transistor OTA: (a) frequency response; (b) rising SR; and (c) input-referred noise.

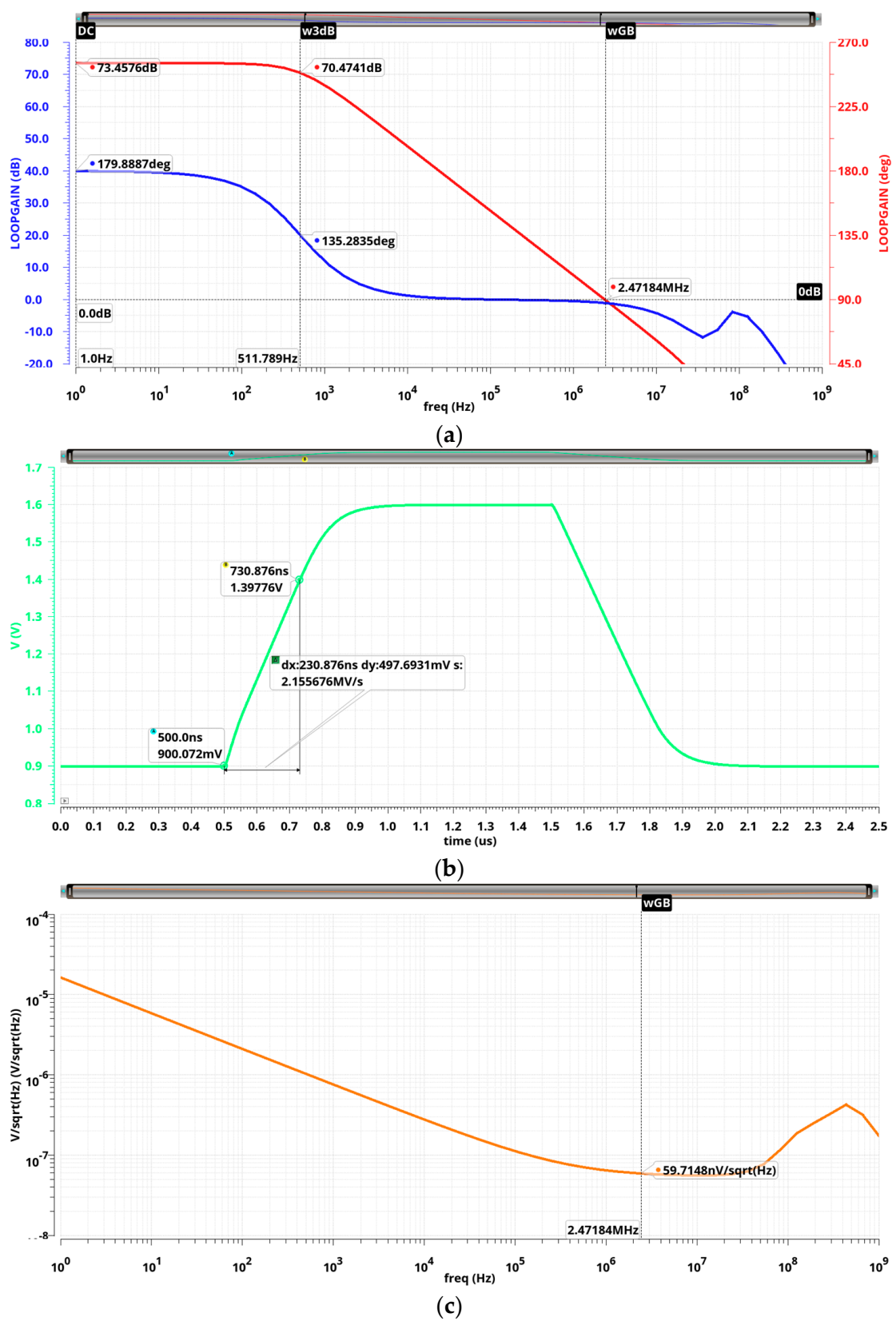


Figure 7. Simulation results of telescopic cascode OTA: (a) frequency response; (b) rising SR; and (c) input-referred noise.

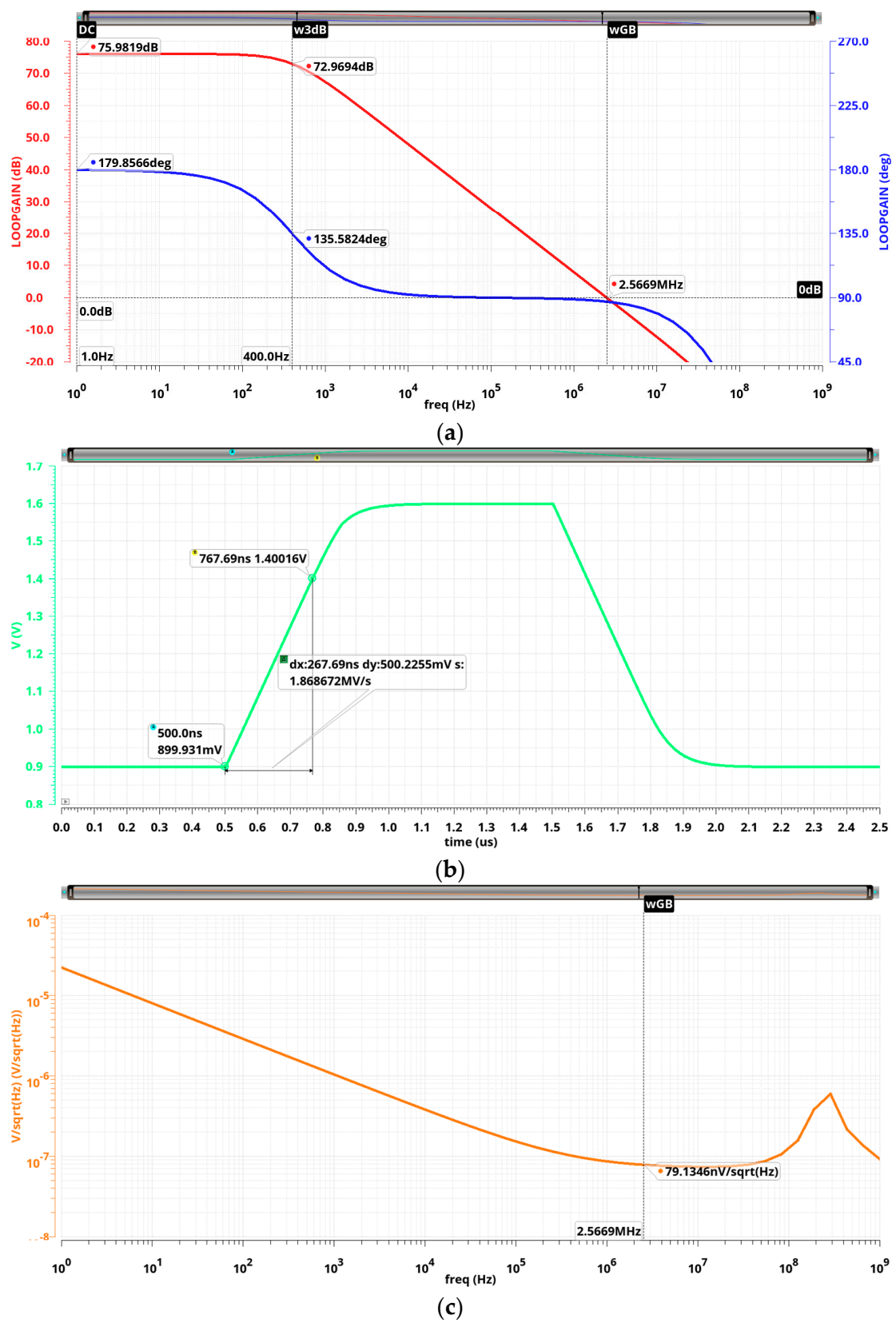


Figure 8. Simulation results of folded cascode OTA: (a) frequency response; (b) rising SR; and (c) input-referred noise.

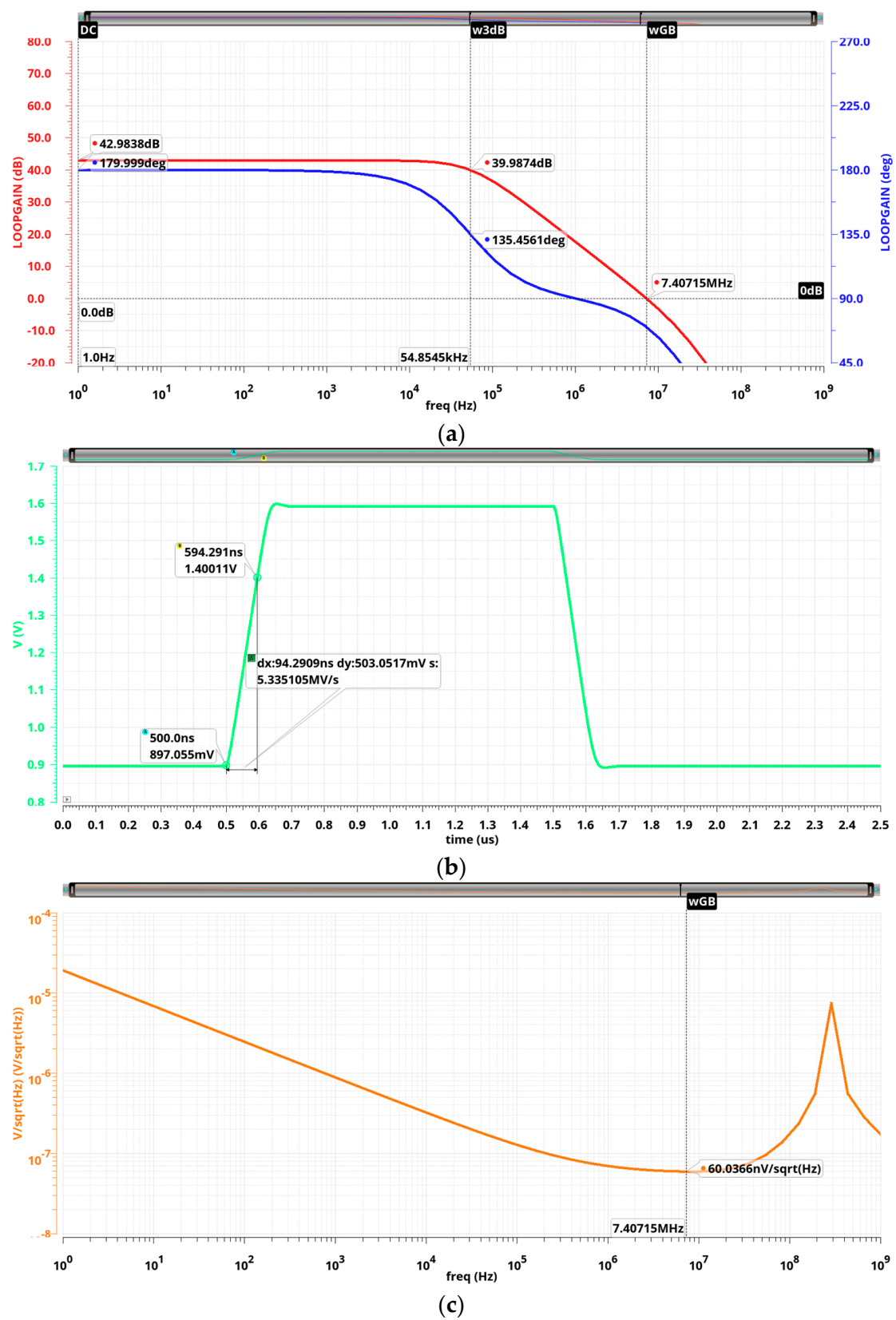


Figure 9. Simulation results of current mirror OTA: (a) frequency response; (b) rising SR; and (c) input-referred noise.

Table 3 summarizes the simulated performance metrics of four types of single-stage OTAs. The folded cascode OTA consumes two times more power than five-transistor and telescopic cascode OTAs due to additional current sources when $I_{D3} = I_{D4} = I_{D5}$. The K of a current mirror OTA is set to 3, which results in a power consumption that is four times higher.

Table 3. Summary of simulated performance metrics.

	5-Transistor OTA	Telescopic Cascode OTA	Folded Cascode OTA	Current Mirror OTA
Current bias	$I_{D5} = 2 \mu\text{A}$	$I_{D5} = 2 \mu\text{A}$	$I_{D3} = I_{D4} = 2 \mu\text{A}$ $I_{D5} = 2 \mu\text{A}$	$I_{D5} = 2 \mu\text{A}$ ($K = 3$)
Power dissipation	3.6 μW	3.6 μW	7.2 μW	14.4 μW
A_o	42.2951 dB	73.4576 dB	75.9819 dB	42.9838 dB
ω_{3dB}	$2\pi \cdot 20$ kHz	$2\pi \cdot 511.79$ Hz	$2\pi \cdot 400$ Hz	$2\pi \cdot 54.8545$ kHz
ω_{GB}	$2\pi \cdot 2.642$ MHz	$2\pi \cdot 2.472$ MHz	$2\pi \cdot 2.567$ MHz	$2\pi \cdot 7.407$ MHz
SR_R	2.028 MV/s	2.156 MV/s	1.868 MV/s	5.335 MV/s
SR_F	1.944 MV/s	1.8469 MV/s	1.879 MV/s	5.499 MV/s
Integrated input-referred noise	0.110 mV _{rms}	0.125 mV _{rms}	0.181 mV _{rms}	0.200 mV _{rms}

Compared to a five-transistor OTA and current mirror OTA, two cascode OTAs achieve much a larger A_o thanks to a higher r_{out} . Here, a five-transistor OTA and current mirror OTA obtain similar A_o values. This is because G_{m,CM_OTA} is K times higher than $G_{m,5TR_OTA}$, but r_{out,CM_OTA} is K times lower than $r_{out,5TR_OTA}$. According to the simulation related to the DC operating condition, r_{out,CM_OTA} is 7.8 Mohm and $r_{out,5TR_OTA}$ is 2.8 Mohm as expected, which results in similar A_o values.

Compared to a five-transistor OTA, two types of cascode OTAs have a smaller ω_{3dB} due to a higher r_{out} while maintaining a similar ω_{GB} . On the contrary, compared to a five-transistor OTA, a current mirror OTA with $K = 3$ improves the ω_{3dB} and ω_{GB} by about three times thanks to the current amplification of additional current mirrors. For the same reason, a current mirror OTA with $K = 3$ improves the SR_R and SR_F by about three times, which are verified by the simulation results.

The integrated input-referred noises of OTAs are compared to effect caused by non-idealities of transistors. The input-referred noises in Figure 6c, Figure 7c, Figure 8c, and Figure 9c are integrated from 1 Hz to $\omega_{GB}/2\pi$. The integrated noise of a telescopic cascode OTA is slightly higher than that of a five-transistor OTA due to cascaded transistors. However, the increase is relatively small because the effect caused by cascoded transistors on the output is negligible. The integrated noise of a folded cascode OTA is higher than that of a telescopic cascode OTA because the effect caused by additional current sources on the output cannot be negligible. A current mirror OTA obtains the largest integrated noise, even with a higher power consumption, due to the current amplification.

5. Discussions and Case Study

Figure 10 presents a comparative analysis of the four types of single-stage OTAs using a spider map, considering both the theoretical and simulated performance metrics summarized in Tables 1 and 3. The five-transistor and telescopic cascode OTAs stand out for their lower power consumption compared to folded cascode and current mirror OTAs with a K of 3. The five-transistor OTA exhibits the simplest structure based on fewer transistors, minimizing the negative effects caused by the nonidealities. The telescopic cascode and folded cascode OTAs achieve a higher r_{out} , resulting in a higher A_o at the sacrifice of swing ranges. While the folded cascode structure improves the input swing

range, its output swing range remains limited like that of the telescopic cascode OTA. On the other hand, the current mirror OTA excels in terms of ω_{GB} , SR_R , and SR_F , but consumes more power. However, it offers the advantage of a wide output swing range even with a relatively small number of transistors at the output node.

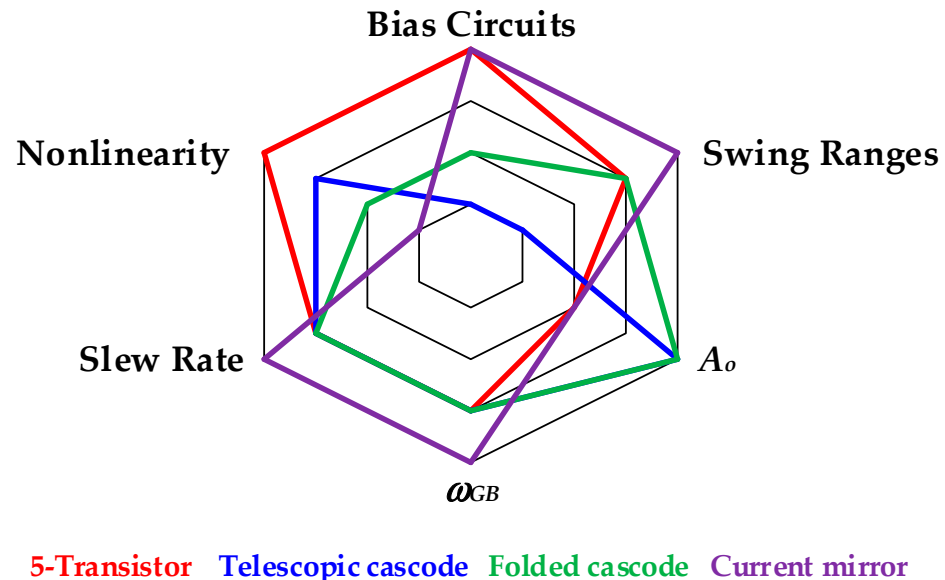


Figure 10. Spider map of four types of single-stage OTAs to compare performance metrics.

The analysis results provide valuable guidance in choosing a suitable OTA structure for specific applications. The five-transistor OTA has the simplest structure, which makes the design easy. It can be employed as a feedforward amplifier in a closed-loop system, but it has been employed as the first stage of multi-stage OP-AMPs [26–29]. Despite its simplicity and moderate performance, since it is difficult for the five-transistor OTA to achieve a high A_o , additional stages are adopted to increase the A_o . The telescopic cascode OTA is a good choice when additional DC current paths are challenging to implement due to a limited power budget, but a high A_o is required. Similar structures based on cascode output stages without additional current sources have been effectively used as feedforward amplifiers in low-power instrumentation amplifiers (IAs) [30] and unity-gain buffers in biomedical applications [31]. The folded cascode OTA has been adopted as the feedforward amplifier of IAs in biopotential recording ICs and sensor interface ICs [32–35] and as an error amplifier of low-dropout regulators (LDOs) [16,17] to utilize a high A_o and a wide input swing range. Compared to telescopic cascode OTAs, using folded cascode OTAs may require a more generous power budget for the system. Moreover, in order to further increase the A_o , the folded cascode structure has been adopted as the first stage of a multi-stage OP-AMP [36]. Lastly, the current mirror OTA has been used as the error amplifier of DC-DC converters and LDOs [18–20] or as the feedforward amplifier of neural recording amplifiers [13,14]. In power converters, the error amplifier often needs to be fast since the output voltage or the load condition can be changed rapidly. In this case, the current mirror OTA could be more suitable than other structures.

At last, a case study is provided to demonstrate the practical application of the tutorial's findings. An appropriate OTA structure is selected based on the analysis results, thereby improving the performance of the conventional neural recording amplifier. In [13], a neural recording amplifier was designed based on a CCIA with the target gain of 40 dB. Here, the current mirror OTA is used as the feedforward amplifier of a CCIA. For the case study, we designed a current mirror OTA based on the transistor sizes in Table 2, and then designed a CCIA based on the current mirror OTA.

Figure 11a shows the closed-loop and open-loop gains of a CCIA based on a current mirror OTA. The closed-loop gain does not reach 40 dB in the mid-band range due to the

low A_o . To address this, we applied the insights from our previous analysis and utilized a telescopic cascode OTA, which offers a significantly higher A_o with a much lower power consumption. Figure 11b shows the closed-loop and open-loop gains of a CCIA based on a telescopic cascode OTA, achieving nearly a 40 dB closed-loop gain in the mid-band range thanks to the higher A_o . Additionally, the current consumption of the telescopic cascode OTA is only one-fourth of that of the current mirror OTA with a K of 3. Furthermore, the integrated input-referred noise of a CCIA based on a current mirror OTA is $78 \mu V_{rms}$, while a CCIA based on a telescopic cascode OTA achieves $66 \mu V_{rms}$ of the integrated input-referred noise. The noise is integrated from 1 Hz to 10 kHz. This indicates that the noise, which is one of important parameters of neural recording amplifiers, can also be improved by adopting a telescopic cascode OTA. In [13], in order to increase the A_o , a cascode output stage was employed in a current mirror OTA. In this structure, the A_o can be increased to as much as the A_o of a telescopic cascode OTA, but this structure would not lead to an improved power consumption and noise compared to a telescopic cascode OTA.

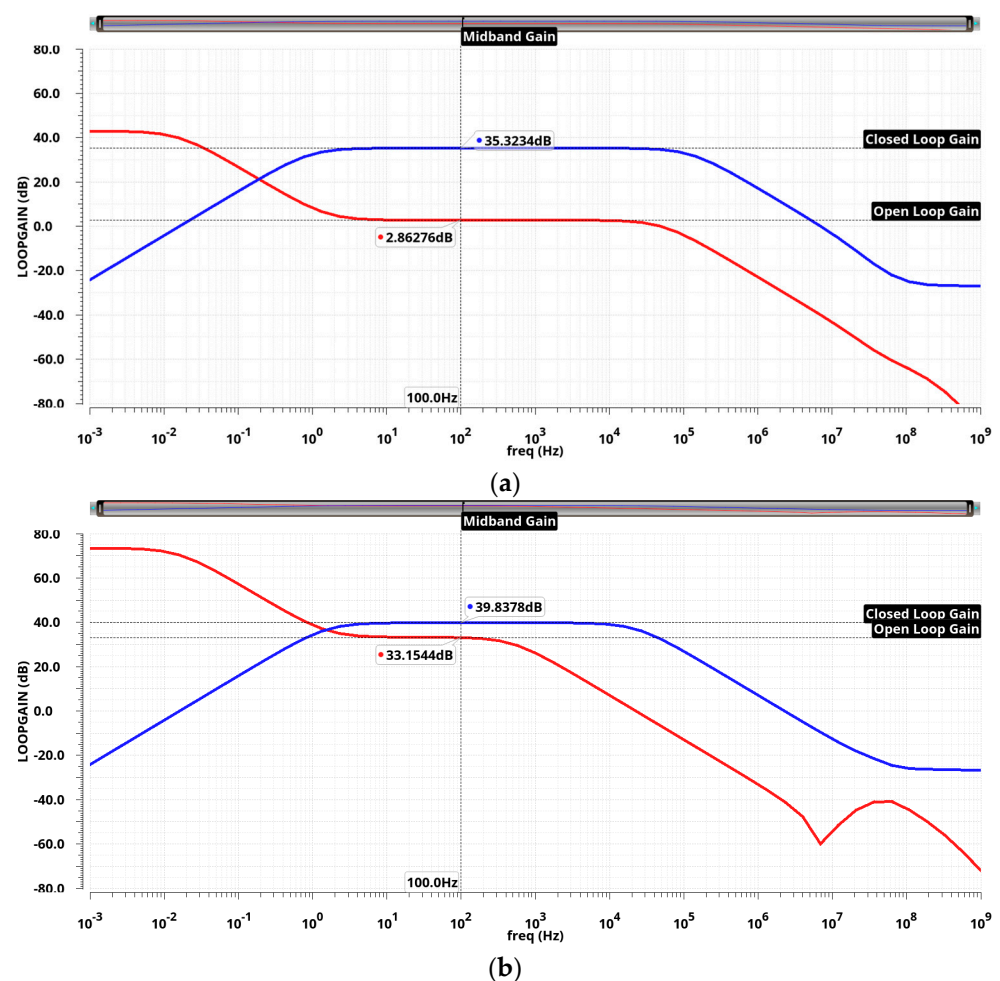


Figure 11. Closed-loop and open-loop gains of CCIA based on (a) current mirror OTA and (b) telescopic cascode OTA. Two OTAs are designed based on transistor sizes shown in Table 2.

The operation principle, key performance metrics, and practical applications of each OTA have been explored with a case study. The performance comparison based on the measurement results can become a promising future work. Moreover, exploring other advanced structures such as current-reusing OTAs [37] and recycling folded cascode OTAs [33] based on this design tutorial can also become a promising and valuable future work.

6. Conclusions

This paper offers a comprehensive design tutorial for four different types of single-stage OTAs: five-transistor OTAs, telescopic cascode OTAs, folded cascode OTAs, and current mirror OTAs. The performance metrics of each OTA are derived through a hand analysis and verified through a simulation. Among the four types of single-stage OTAs, the five-transistor OTA stands out for its simple structure, minimizing the negative effects of the nonidealities by reducing the number of transistors. On the other hand, telescopic cascode and folded cascode OTAs achieve higher r_{out} and A_o at the expense of swing ranges. The input swing range can be improved by the folded cascode structure at the expense of power consumption, but the output swing range of a folded cascode OTA is still limited like that of a telescopic cascode OTA. A current mirror OTA excels in terms of bandwidth and SRs but consumes more power. Additionally, it offers a wide output swing range thanks to the small number of transistors at the output node. The insights gained from this tutorial will serve as a valuable resource for designing high-performance analog amplifiers in diverse applications.

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